Exercise 1: Setting Up JUnit

Add JUnit Dependency

If using Maven, add this to `pom.xml`.

```
<dependencies>

  <dependency>
     <groupId>junit</groupId>
     <artifactId>junit</artifactId>
     <version>4.13.2</version>
     <scope>test</scope>
  </dependency>

</dependencies>
```

Create a Test Class

```
import org.junit.Test;

public class CalculatorTest {

  @Test
  public void testSample() {
     System.out.println("JUnit is working!");
  }
}
```

Exercise 2: Writing Basic JUnit Tests

Calculator.java

```
public class Calculator {

  public int add(int a, int b) {
     return a + b;
  }
```

```java
    public int subtract(int a, int b) {
        return a - b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }

    public int divide(int a, int b) {
        return a / b;
    }
}
```

CalculatorTest.java

```java
import static org.junit.Assert.assertEquals;
import org.junit.Test;

public class CalculatorTest {

    Calculator calculator = new Calculator();

    @Test
    public void testAddition() {
        assertEquals(10, calculator.add(4, 6));
    }

    @Test
    public void testSubtraction() {
        assertEquals(5, calculator.subtract(10, 5));
    }

    @Test
    public void testMultiplication() {
        assertEquals(20, calculator.multiply(4, 5));
    }

    @Test
    public void testDivision() {
        assertEquals(2, calculator.divide(10, 5));
    }
}
```

Exercise 3: Assertions in JUnit

```
import static org.junit.Assert.*;
import org.junit.Test;

public class AssertionsTest {

    @Test
    public void testAssertions() {

        // Assert Equals
        assertEquals(5, 2 + 3);

        // Assert True
        assertTrue(10 > 5);

        // Assert False
        assertFalse(10 < 5);

        // Assert Null
        String name = null;
        assertNull(name);

        // Assert Not Null
        Object obj = new Object();
        assertNotNull(obj);

        // Assert Same
        String s1 = "Java";
        String s2 = s1;
        assertSame(s1, s2);

        // Assert Not Same
        String s3 = new String("Java");
        assertNotSame(s1, s3);

        // Assert Array Equals
        int[] expected = {1, 2, 3};
        int[] actual = {1, 2, 3};
        assertArrayEquals(expected, actual);
    }
}
```

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown

Calculator.java

```
public class Calculator {

  public int add(int a, int b) {
     return a + b;
  }
}
```

CalculatorTest.java

```
import static org.junit.Assert.assertEquals;

import org.junit.After;
import org.junit.Before;
import org.junit.Test;

public class CalculatorTest {

  private Calculator calculator;

  @Before
  public void setUp() {
     System.out.println("Setting up...");
     calculator = new Calculator();
  }

  @After
  public void tearDown() {
     System.out.println("Cleaning up...");
     calculator = null;
  }

  @Test
```

```java
    public void testAdd() {

        // Arrange
        int a = 15;
        int b = 5;

        // Act
        int result = calculator.add(a, b);

        // Assert
        assertEquals(20, result);
    }
}
```

Output

Setting up...
Cleaning up...

If the test passes:

Tests run: 1
Failures: 0
Errors: 0